ModelIt!

Teacher's Guide

The Food Web Puzzle

Remove One Piece and Watch Everything Change

Quick Reference

Grade Level	Grade 05L2
NGSS Standard	5-LS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-LS2-1

Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing matter cycling through multiple trophic levels in a food web.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems** — Students model how organisms depend on each other for matter and energy.
Crosscutting Concept	**Systems and System Models** — Students model the food web as an interconnected system where changes cascade through all levels.

CAST Testing Alignment

MULTIPLE CHOICE:

In a grassland, wolves eat rabbits and rabbits eat grass. If wolves are removed, what happens first?

CONSTRUCTED RESPONSE:

Explain how removing one species from a food web can cause a trophic cascade, using evidence from your model simulation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

A food web shows the complex feeding relationships in an ecosystem. Unlike a simple food chain, a food web reveals how organisms are interconnected — and how removing one piece can cause a cascade of effects.

Producers: The Foundation

Plants and algae are producers that capture sunlight energy through photosynthesis. They form the base of every food web and provide energy for all consumers above.

Consumers: Energy Transfers

Primary consumers (herbivores) eat producers. Secondary consumers (predators) eat herbivores. At each level, about 90% of energy is lost as heat, which is why predators are always rarer than prey.

Trophic Cascades

When a species is removed, effects cascade through the web. The reintroduction of wolves to Yellowstone in 1995 is a famous example — wolves reduced elk overpopulation, allowing trees and rivers to recover.

Decomposers Close the Loop

Bacteria, fungi, and invertebrates break down dead organisms, releasing nutrients back into the soil. Without decomposers, nutrients would be locked in dead matter and producers would starve.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Producer Population, Primary Consumer Population, Secondary Consumer Population, Decomposer Activity, and Available Nutrients as the five system components.

E **Establish Relationships**

Students determine that Producers feed Primary Consumers (+), Primary Consumers feed Secondary Consumers (+), Secondary Consumers reduce Primary Consumers (-), Decomposers release Nutrients (+), and Nutrients support Producers (+).

V **Visualize & Model**

Students build the food web model showing circular matter flow from producers through consumers to decomposers and back to nutrients for producers.

E **Evaluate Outcomes**

Students run the Remove Predators scenario and observe herbivore explosion followed by producer decline, then run Remove Producers to see total collapse.

R **Revise & Extend**

Students add Sunlight Energy, Invasive Species, or a Tertiary Consumer to make the model more complex and realistic.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What happens when you remove one puzzle piece from nature?"*

Watch For

Ask: If all rabbits disappeared, what would happen? What about all plants?

Slide 2: Learning Objectives

2 min

Say: *"Today we build a food web and test what happens when species disappear."*

Watch For

Students read objectives and identify vocabulary they already know.

Slide 3: Big Question

2 min

Say: *"This is called a trophic cascade, and it has happened in real life."*

Watch For

Share the Yellowstone wolves example briefly.

Slide 4: LEVER Framework

2 min

Say: *"Our model has a CIRCULAR flow — nutrients come back around to the beginning."*

Watch For

Preview how decomposers feed nutrients back to producers.

Slide 5: Activity 1: Components

8 min

Say: *"Producer Population is our only external component. Everything else responds to it."*

Watch For

Students sort components into external and internal, discussing why.

Slide 6: Activity 2: Connections

8 min

Say: *"Notice the circle: producers feed consumers, dead consumers feed decomposers, decomposers feed nutrients back to producers."*

Watch For

Students draw all five relationship arrows.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens when we remove all predators? Let's watch."*

Watch For

Students run Remove Predators and Remove Producers scenarios.

Slide 8: Discoveries

5 min

Say: *"Everything is connected. Remove one piece and the whole system shifts."*

Watch For

Class discussion: Which removal caused the biggest surprise?

Slide 9: STEM Challenge

10 min

Say: *"A lake lost its top predator. Your job: restore the balance."*

Watch For

Groups design restoration plans; present to class.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Producer Population
- **INTERNAL:** Primary Consumer Population, Secondary Consumer Population, Decomposer Activity, Available Nutrients

Relationships

- **Producer Population to Primary Consumer Population:** POSITIVE (+) — More plants provide more food for herbivores, supporting larger populations.
- **Primary Consumer Population to Secondary Consumer Population:** POSITIVE (+) — More herbivores provide more food for predators.
- **Secondary Consumer Population to Primary Consumer Population:** NEGATIVE (-) — More predators eat more herbivores, reducing their population.
- **Decomposer Activity to Available Nutrients:** POSITIVE (+) — More decomposition releases more nutrients into the soil.
- **Available Nutrients to Producer Population:** POSITIVE (+) — More soil nutrients help producers grow, supporting larger plant populations.

Key Discoveries

- Every species in a food web is connected — removing one causes a cascade of changes
- Without predators, herbivores overpopulate and can destroy producers through overgrazing
- Without producers, the entire food web collapses because there is no energy entering the system
- Decomposers close the loop by recycling dead matter back into nutrients for producers

Multiple Choice Post-Assessment Answer Key

1. In the food web model, what happens when Secondary Consumers (predators) are removed from the system?

Correct: B) Primary Consumers explode in population, overeat the Producers, and then Producers crash

Correct! Without predators controlling their numbers, herbivores (Primary Consumers) overpopulate. They consume Producers faster than plants can regrow, causing a Producer crash. This chain reaction is called a trophic cascade.

2. What happens in the model when Producer Population is locked at 0%?

Correct: C) The entire food web collapses because no energy enters the system

Exactly! Producers are the foundation of the food web. They capture sunlight energy and make it available to all other organisms. Without producers, primary consumers starve, then secondary consumers starve, and the entire ecosystem collapses.

3. What role do decomposers play in completing the food web cycle?

Correct: B) They break down dead organisms and release nutrients back into the soil for producers to use

That's right! Decomposers break down dead plants and animals, releasing nutrients into the soil. Producers then absorb those nutrients to grow. This closes the loop, making the food web a true cycle rather than a dead end.

4. In a grassland, wolves eat rabbits and rabbits eat grass. If wolves are removed, what happens first?

Correct: B) Rabbit population increases rapidly because nothing is eating them

Correct! Wolves are the secondary consumers that control rabbit (primary consumer) populations. Removing wolves means rabbits face no predation, so their population explodes. Eventually, the overabundant rabbits would overgraze the grass.

5. Why do ecologists say 'everything is connected' in an ecosystem?

Correct: B) Because changing one species population causes cascading effects on every other species through the food web

Exactly! The model demonstrates that removing or changing one component affects every other component through their relationships. Producers support consumers, consumers control each other, and decomposers recycle nutrients for producers. Every connection matters.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Food chains are straight lines"	Real ecosystems are food WEBS with many interconnected chains. An animal may eat multiple food sources and be eaten by multiple predators.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Removing predators helps prey animals"	In the short term, prey populations increase. But without predators controlling their numbers, prey overpopulate, destroy their food source, and eventually crash.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Decomposers are not important"	Without decomposers, dead matter would pile up and nutrients would never return to the soil. Producers would run out of nutrients and the entire food web would collapse.